

2_10xMouse_PBMC_STARsolo

August 31, 2026

```
[1]: library(Seurat)
library(ggplot2)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
getwd()
dir.create("figures_10xMouse_PBMC")
dir.create("data_10xMouse_PBMC")
```

Loading required package: SeuratObject

Loading required package: sp

‘SeuratObject’ was built under R 4.3.3 but the current version is 4.4.1; it is recommended that you reinstall ‘SeuratObject’ as the ABI for R may have changed

‘SeuratObject’ was built with package ‘Matrix’ 1.6.4 but the current version is 1.7.5; it is recommended that you reinstall ‘SeuratObject’ as the ABI for ‘Matrix’ may have changed

Attaching package: ‘SeuratObject’

The following objects are masked from ‘package:base’:

intersect, t

Warning message:

"package ‘dplyr’ was built under R version 4.4.3"

Attaching package: ‘dplyr’

The following objects are masked from ‘package:stats’:

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

Warning message:

"package 'RColorBrewer' was built under R version 4.4.3"

"/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'

Warning message in dir.create("figures_10xMouse_PBMC"):

"'figures_10xMouse_PBMC' already exists"

Warning message in dir.create("data_10xMouse_PBMC"):

"'data_10xMouse_PBMC' already exists"

```
[6]: # celltype annotation

celltypes <- read.table(paste0("data_", dataset_id, "/celltype_annotation.tsv"),
  ↪ sep = "\t")
colnames(celltypes) <- c("barcode", "celltype")
rownames(celltypes) <- celltypes$barcode

PBMCelltypeColors <- c("B_cells"="#6D5A5D",
  "Dendritic Cells"="#C4B3AB", "#E78063", "#c492a7",
  "Monocytes"="#81B29A",
  "Platelets"="#e78063",
  "T_cells"="#84B6D6",
  "Natural Killers"= "#F2CC8F",
  "Neutrophils"="#c492a7",
  "Unknown" = "#3D405B")
```

```
[8]: dataset_id <- "10xMouse_PBMC"
sample <- "10k"

path <- paste0("/mnt/TEresults/snakefile_results/results/STARsolo_EM_TE/
  ↪ ", dataset_id, "/", sample, "/TE_Solo.out/Gene/")

# get filtered barcodes
STAR_path <- paste0("/mnt/TEresults/snakefile_results/results/STARoutdir/
  ↪ ", dataset_id, "/", sample, "/best_Solo.out/Gene")
filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1
  ↪ # read barcodes selected by STARsolo
# remove cells annotated as multipllets by 10x
```

```

mouse_assignment <- read.csv("data_10xMouse_PBMC/
  ↪SC3_v3_NextGem_DI_CellPlex_Mouse_PBMC_10K_Multiplex_multiplexing_analysis_assignment_confid
  ↪csv")
mouse_assignment$Barcodes <- sapply(strsplit(mouse_assignment$Barcodes,
  ↪split="-"), "[", 1)
multiplets <-
  ↪mouse_assignment$Barcodes[mouse_assignment$Assignment=="Multiplet"]
filteredBarcodes <- setdiff(filteredBarcodes, multiplets)

STARsolo_TE_EM_mat <- Seurat::ReadMtx(
  mtx = paste0(path, "/raw/UniqueAndMult-EM.mtx"),
  cells = paste0(path, "/raw/barcodes.tsv"),
  features = paste0(path, "/raw/features.tsv")
) # read matrix
STARsolo_TE_EM_mat <- STARsolo_TE_EM_mat[, filteredBarcodes] # keep only
  ↪barcodes passing thresholds

nCells <- length(filteredBarcodes)
thrMinCells <- round(nCells * 0.05)
thrMinCells

objTE_STARsolo <- Seurat::CreateSeuratObject(STARsolo_TE_EM_mat,
  project = "STARsolo_TE_EM",
  min.cells = thrMinCells, min.features = 0
)

objTE_STARsolo

```

240

Warning message:

"Feature names cannot have underscores ('_'), replacing with dashes ('-')"

An object of class Seurat

3232 features across 4790 samples within 1 assay

Active assay: RNA (3232 features, 0 variable features)

1 layer present: counts

```

[9]: annotation_stellarscope <- read.table("annotation/annotation_stellarscope.tsv")
  ↪# generated with annotation_scripts/create_annotations_mouse.Rmd
head(annotation_stellarscope)

```

		chr	start	end	sw_score	strand	locusID	family	subfam
		<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<chr>
A data.frame: 6 × 10	1	chr1	3000001	3002128	12955	-	L1_Mus3_dup24	L1	L1-Mu
	2	chr1	3003153	3003994	1216	-	L1Md_F_dup2	L1	L1Md-
	3	chr1	3003994	3004054	234	-	L1_Mus3_dup25	L1	L1-Mu
	4	chr1	3004041	3004206	3685	+	L1_Rod_dup1	L1	L1-Ro
	5	chr1	3004271	3005001	3685	+	L1_Rod_dup2	L1	L1-Ro
	6	chr1	3005002	3005439	1280	+	L1_Rod_dup3	L1	L1-Ro

```
[10]: # Remove some classes of TEs (already not present in SoloTE)
classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")
starsoloTEs <- Features(objTE_STARsolo)

filteredStarsoloTEs <-
  ↪ annotation_stellarscope[annotation_stellarscope$stellarscopeID %in%
  ↪ starsoloTEs, ] %>%
  dplyr::filter(!class %in% classesToExclude) %>%
  pull(stellarscopeID)

print("Percentage of TEs in removed orders:")
print(length(setdiff(starsoloTEs, filteredStarsoloTEs)) / length(starsoloTEs) *
  ↪ 100)

objTE_STARsolo <- objTE_STARsolo[intersect(starsoloTEs, filteredStarsoloTEs), ]

print("Summary of n counts")
print(summary(objTE_STARsolo$nCount_RNA))
print("Summary of n feature")
print(summary(objTE_STARsolo$nFeature_RNA))

print("Loci present in the annotation:")
print(table(rownames(objTE_STARsolo) %in%
  ↪ annotation_stellarscope$stellarscopeID))
```

```
[1] "Percentage of TEs in removed orders:"
[1] 1.54703
[1] "Summary of n counts"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
9.562 344.061 445.935 534.795 646.879 2364.368
[1] "Summary of n feature"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
17.0   291.0   353.0   368.2  431.0   888.0
[1] "Loci present in the annotation:"
```

```
TRUE
3182
```

```
[11]: feature_metadata <- annotation_stellarscope[match(Features(objTE_STARsolo),
  ↪annotation_stellarscope$stellarscopeID),]

head(feature_metadata)
```

		chr	start	end	sw_score	strand	locusID	
		<chr>	<int>	<int>	<int>	<chr>	<chr>	<
	309	chr1	3274149	3274468	2745	+	ERVB7_1-LTR_MM_dup2	fa
	314	chr1	3280116	3280435	2733	+	ERVB7_1-LTR_MM_dup3	E
A data.frame: 6 × 10	1761	chr1	4556988	4557455	2336	-	IAPLTR2_Mm	E
	1767	chr1	4559115	4559597	3711	-	IAPEz-int_dup8	E
	1771	chr1	4567176	4567643	2336	-	IAPLTR2_Mm_dup1	E
	2675	chr1	5102985	5103159	285	-	MIRb_dup22	M

```
[12]: options(repr.plot.width=7, repr.plot.height=5)

objTE_STARsolo@meta.data$nCount_TE <- objTE_STARsolo@meta.data$nCount_RNA
objTE_STARsolo@meta.data$nFeature_TE <- objTE_STARsolo@meta.data$nFeature_RNA
# Visualize QC metrics as a violin plot
VlnPlot(objTE_STARsolo, features = c("nCount_TE"), ncol = 1,
  pt.size = 0) + theme(text=element_text(size=17))
ggsave(paste0("figures_", dataset_id, "/nCountTE_violin_STARsolo.pdf"),
  ↪device='pdf')

VlnPlot(objTE_STARsolo, features = c("nFeature_TE"), ncol = 1,
  pt.size = 0) + theme(text=element_text(size=17))
ggsave(paste0("figures_", dataset_id, "/nFeatureTE_violin_STARsolo.pdf"),
  ↪device='pdf')
```

Warning message:

"Default search for "data" layer in "RNA" assay yielded no results; utilizing "counts" layer instead."

Warning message:

"The `slot` argument of `FetchData()` is deprecated as of SeuratObject 5.0.0.

Please use the `layer` argument instead.

The deprecated feature was likely used in the [Seurat](#) package.

Please report the issue at

<https://github.com/satijalab/seurat/issues>."

Warning message:

"`PackageCheck()` was deprecated in SeuratObject 5.0.0.

Please use `rlang::check\_installed()` instead.

The deprecated feature was likely used in the [Seurat](#) package.

Please report the issue at

```
<https://github.com/satijalab/seurat/issues>."
```

Warning message:

```
"`aes_string()` was deprecated in ggplot2 3.0.0.
```

```
  Please use tidy evaluation idioms with `aes()`.
  See also `vignette("ggplot2-in-packages")` for more information.
```

```
  The deprecated feature was likely used in the Seurat
```

```
package.
```

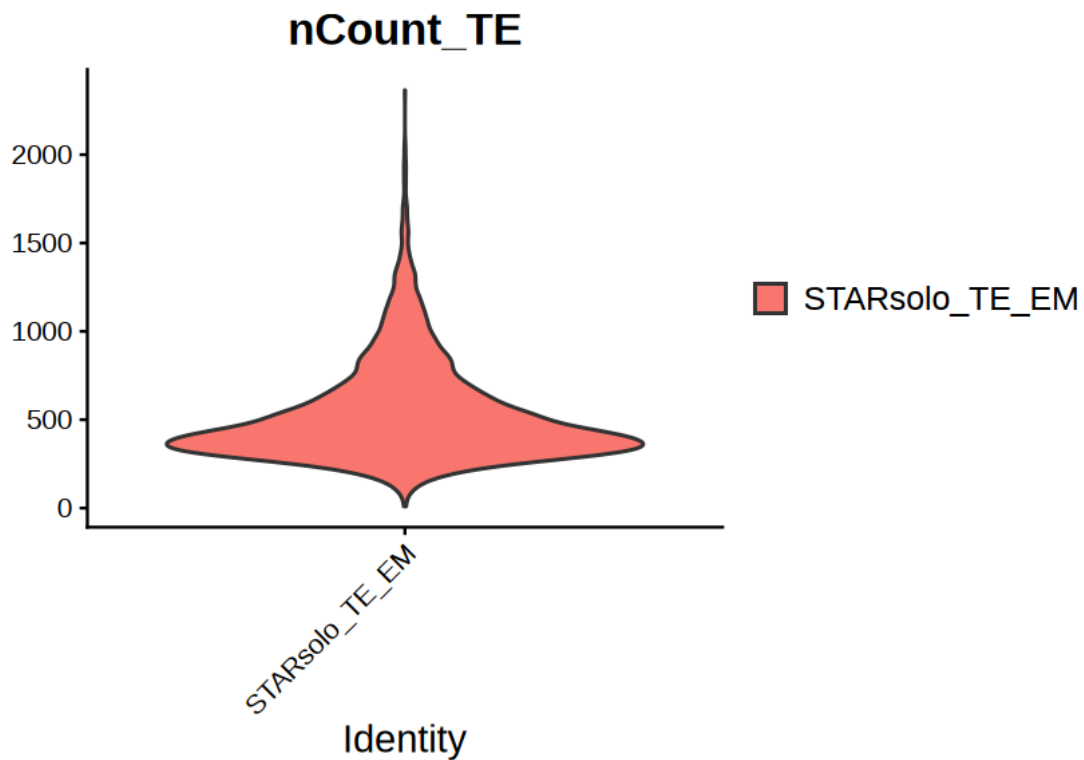
```
  Please report the issue at
```

```
<https://github.com/satijalab/seurat/issues>."
```

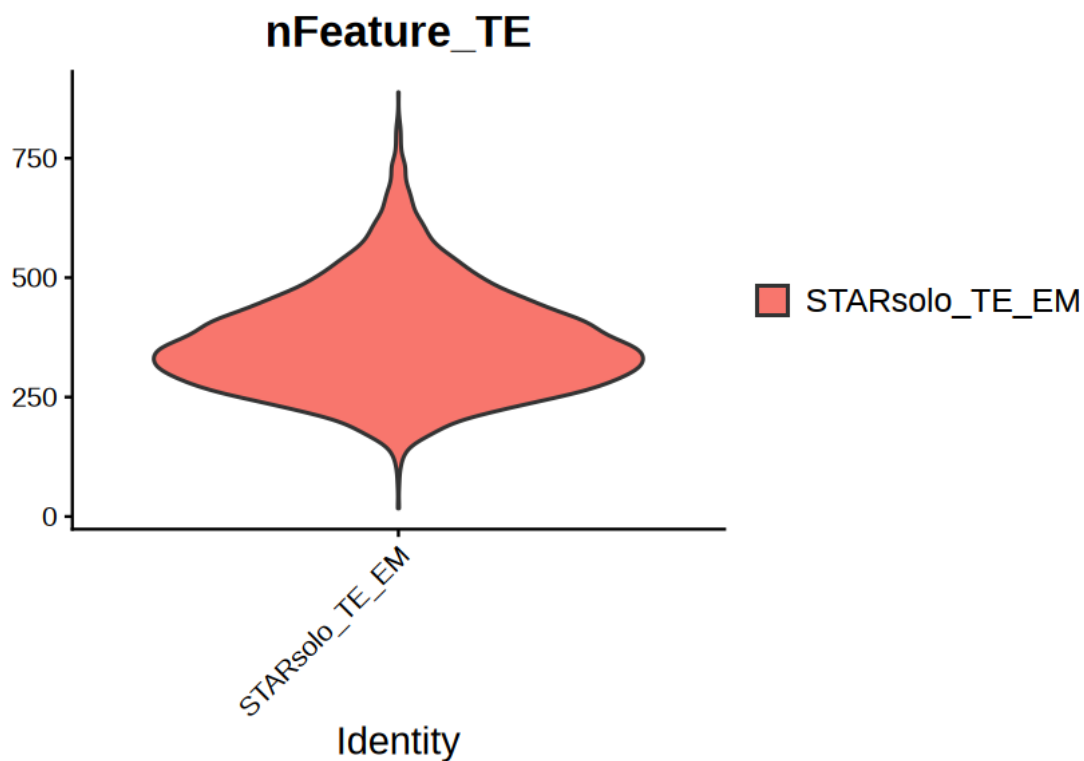
```
Saving 6.67 x 6.67 in image
```

Warning message:

```
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."
```



```
Saving 6.67 x 6.67 in image
```



```
[13]: objTE_STARsolo <- JoinLayers(objTE_STARsolo)

objTE_STARsolo <- NormalizeData(objTE_STARsolo, normalization.method = "
  ↪LogNormalize", scale.factor = 10000)

gc()

objTE_STARsolo <- FindVariableFeatures(objTE_STARsolo, selection.method = "
  ↪vst", nfeatures = 4000)

gc()

# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objTE_STARsolo), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objTE_STARsolo)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

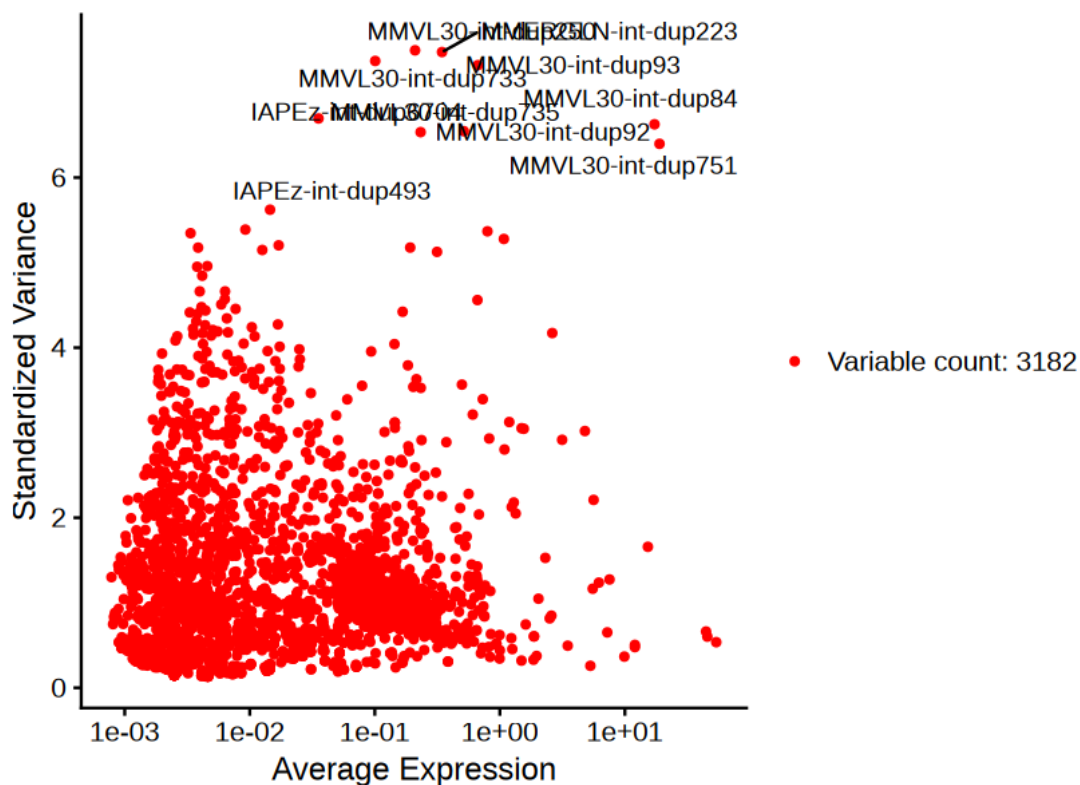
Normalizing layer: counts

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2×6 of type dbl	Ncells	15120095	807.6	39072294	2086.7	34156867	1824.2
	Vcells	112593584	859.1	204769736	1562.3	212369191	1620.3

Finding variable features for layer counts

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2×6 of type dbl	Ncells	15130569	808.1	39072294	2086.7	34156867	1824.2
	Vcells	112638608	859.4	204769736	1562.3	212369191	1620.3

When using `repel`, set `xnudge` and `ynudge` to 0 for optimal results



```
[14]: gc()
all.genes <- rownames(objTE_STARsolo)
objTE_STARsolo <- ScaleData(objTE_STARsolo) # on hvg
```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2×6 of type dbl	Ncells	15152845	809.3	39072294	2086.7	34156867	1824.2
	Vcells	108654700	829.0	204769736	1562.3	212369191	1620.3

Centering and scaling data matrix


```
[15]: grep("ERVVL", all.genes, value = T)[1:20]
```

1. 'MERVL-int-dup2537' 2. NA 3. NA 4. NA 5. NA 6. NA 7. NA 8. NA 9. NA 10. NA 11. NA
12. NA 13. NA 14. NA 15. NA 16. NA 17. NA 18. NA 19. NA 20. NA

```
[16]: objTE_STARsolo <- RunPCA(objTE_STARsolo, features = VariableFeatures(object =  
  ↪objTE_STARsolo))  
  
DimPlot(objTE_STARsolo, reduction = "pca") + NoLegend()  
  
ElbowPlot(objTE_STARsolo)
```

PC_ 1

Positive: RMER5-dup4769, RLTR6-Mm-dup166, MMVL30-int-dup751, RSINE1-dup7030,
RMER20B-dup1279, RLTR6-Mm-dup14, RLTR6-Mm-dup13, B3-dup44048, RSINE1-dup110789,
L2c-dup11353

MMVL30-int-dup84, MLT1B-dup8769, L1-Mus3-dup16098, Lx-dup23147,
ERVB7-2-LTR-MM-dup301, B1-Mur1-dup2948, MTD-dup36599, Lx7-dup13295,
L1Md-F3-dup5431, L1Md-F2-dup5378

MLT1A1-dup1566, RMER15-dup10472, B1-Mur2-dup8227, B1F-dup13336,
MMVL30-int-dup478, B2-Mm2-dup1895, B3-dup79838, L1-Mus1-dup19250, ERVB7-2-LTR-
MM-dup302, L1Md-F3-dup631

Negative: MMVL30-int-dup685, RLTR6-Mm-dup145, RLTR6C-Mm-dup63, RLTR4-MM-int-
dup287, L1Md-F3-dup14136, RLTR4-MM-int-dup288, ERVB4-1C-LTR-Mm-dup282,
L1-Mus1-dup14975, B1-Mus1-dup50841, B1-Mus1-dup50811

B1-Mus1-dup50835, B1-Mus1-dup50799, B1-Mus1-dup50776,
B1-Mus1-dup50752, L1M2-dup15802, B1-Mus1-dup92817, L1-Mus3-dup46173, MMVL30-int-
dup885, RLTR4-Mm-dup170, RLTR4-Mm-dup171

Lx9-dup46147, MT2B1-dup2418, B1-Mus1-dup50838, B1-Mus1-dup50779,
B1-Mus1-dup50761, B1-Mus1-dup50832, B1-Mus1-dup50814, B1-Mus1-dup50773,
B1-Mus1-dup50749, B1-Mus1-dup50737

PC_ 2

Positive: MMVL30-int-dup943, L1Md-F3-dup14136, L1M2-dup15802, B1-Mus1-dup50761,
B1-Mus1-dup50779, B1-Mus1-dup50796, B1-Mus1-dup50802, B1-Mus1-dup50808,
B1-Mus1-dup50737, B1-Mus1-dup50814

B1-Mus1-dup50749, B1-Mus1-dup50755, B1-Mus1-dup50820,
B1-Mus1-dup50838, B1-Mus1-dup50844, B1-Mus1-dup50773, B1-Mus1-dup50832,
B1-Mus1-dup50881, B1-Mus1-dup50767, B1-Mus1-dup50826

B1-Mus1-dup50731, B1-Mus1-dup50790, B1-Mus1-dup50784,
B1-Mus1-dup50850, B1-Mus1-dup50860, B1-Mus1-dup50799, B1-Mus1-dup50835,
B1-Mus1-dup50752, B1-Mus1-dup50811, B1-Mus1-dup50776

Negative: RLTR4-Mm-dup170, RLTR4-Mm-dup171, RLTR4-MM-int-dup288, RLTR4-MM-int-
dup130, RLTR4-MM-int-dup287, Lx3B-dup2078, RLTR53-Mm-dup209, RLTR4-Mm-dup52,
RLTR4-Mm-dup50, RLTR4-Mm-dup26

RLTR4-Mm-dup24, RLTR4-Mm-dup147, RLTR4-Mm-dup151, RLTR4-Mm-dup149,
RLTR4-Mm-dup145, RLTR4-Mm-dup143, RLTR4-Mm-dup215, ORR1C2-dup9583, IAPEz-int-
dup6834, RLTR4-Mm-dup219

L1Md-F2-dup23486, RLTR4-Mm-dup141, RLTR4-Mm-dup96, RLTR4-Mm-dup95,

RLTR4-Mm-dup88, RLTR4-Mm-dup87, RLTR4-Mm-dup36, RLTR4-Mm-dup193, RLTR4-Mm-dup192, RLTR4-Mm-dup74

PC_ 3

Positive: B2-Mm1a-dup5102, B2-Mm1a-dup12896, B2-Mm1a-dup12806, B2-Mm1t-dup12926, B2-Mm1t-dup14884, B2-Mm1a-dup9527, B2-Mm1a-dup7726, B2-Mm1a-dup2700, B2-Mm1a-dup15634, B2-Mm1a-dup8970

B2-Mm1a-dup7582, B2-Mm1a-dup11368, B2-Mm1a-dup17047, B2-Mm1a-dup2256, B2-Mm1t-dup6425, B2-Mm1t-dup3308, B2-Mm1a-dup13001, B2-Mm1t-dup18176, B2-Mm1a-dup16372, B2-Mm1a-dup12636

B2-Mm1t-dup10228, B2-Mm1a-dup15214, B2-Mm1t-dup3482, B2-Mm1a-dup14687, B2-Mm1t-dup12636, B2-Mm1a-dup13612, B2-Mm1a-dup10083, B2-Mm1a-dup3576, B2-Mm1a-dup17735, B2-Mm1a-dup14085

Negative: IAPLTR1-Mm-dup606, IAPLTR1-Mm-dup1003, IAPLTR1-Mm-dup1002, IAPLTR1-Mm-dup645, IAPLTR1-Mm-dup222, IAPLTR1-Mm-dup607, IAPLTR1-Mm-dup221, IAPLTR1-Mm-dup644, IAPLTR1-Mm-dup1428, IAPLTR1-Mm-dup243

IAPLTR1-Mm-dup861, IAPLTR1-Mm-dup244, IAPLTR1-Mm-dup233, IAPLTR1-Mm-dup180, IAPLTR1-Mm-dup234, IAPLTR1-Mm-dup1429, IAPLTR1-Mm-dup179, IAPLTR1-Mm-dup1152, IAPLTR1-Mm-dup1153, IAPLTR1-Mm-dup648

IAPLTR1-Mm-dup649, IAPLTR1-Mm-dup1202, IAPLTR1-Mm-dup627, IAPLTR1-Mm-dup626, IAPLTR1-Mm-dup1201, IAPLTR1-Mm-dup117, IAPLTR1-Mm-dup524, IAPLTR1-Mm-dup525, IAPLTR1-Mm-dup858, IAPLTR1-Mm-dup859

PC_ 4

Positive: RLTR4-MM-int-dup288, RLTR4-MM-int-dup287, Lx9-dup1463, IAPEz-int-dup6834, Lx3B-dup3459, RLTR4-Mm-dup170, RLTR4-Mm-dup171, RLTR53-Mm-dup209, Lx5-dup3006, L1Md-F2-dup57836

RLTR4-Mm-dup215, RLTR4-Mm-dup36, RLTR4-Mm-dup118, RLTR4-Mm-dup117, RLTR4-Mm-dup87, RLTR4-Mm-dup88, RLTR4-Mm-dup96, RLTR4-Mm-dup95, RLTR4-Mm-dup192, RLTR4-Mm-dup193

RLTR4-Mm-dup74, RLTR4-Mm-dup73, RLTR4-Mm-dup76, RLTR4-Mm-dup75, RLTR4-Mm-dup35, RLTR6-Mm-dup145, RLTR6C-Mm-dup63, RLTR4-Mm-dup141, RLTR4-Mm-dup143, IAPEz-int-dup3989

Negative: B2-Mm1a-dup7726, B2-Mm1a-dup12896, B2-Mm1a-dup5102, B2-Mm1a-dup12806, B2-Mm1t-dup14884, B2-Mm1t-dup12926, B2-Mm1a-dup13001, B2-Mm1t-dup3308, B2-Mm1a-dup9527, B2-Mm1a-dup17047

B2-Mm1a-dup2256, B2-Mm1a-dup7582, B2-Mm1a-dup8970, B2-Mm1a-dup11368, B2-Mm1t-dup6425, B2-Mm1a-dup15634, B2-Mm1t-dup3482, B2-Mm1a-dup2700, B2-Mm1t-dup12636, B2-Mm1a-dup13612

B2-Mm1t-dup18176, B2-Mm1a-dup16372, B2-Mm1a-dup12636, B2-Mm1t-dup10228, B2-Mm1a-dup10083, B2-Mm1a-dup14687, B2-Mm1a-dup15214, B2-Mm1t-dup19058, B2-Mm1a-dup14446, B2-Mm1a-dup16358

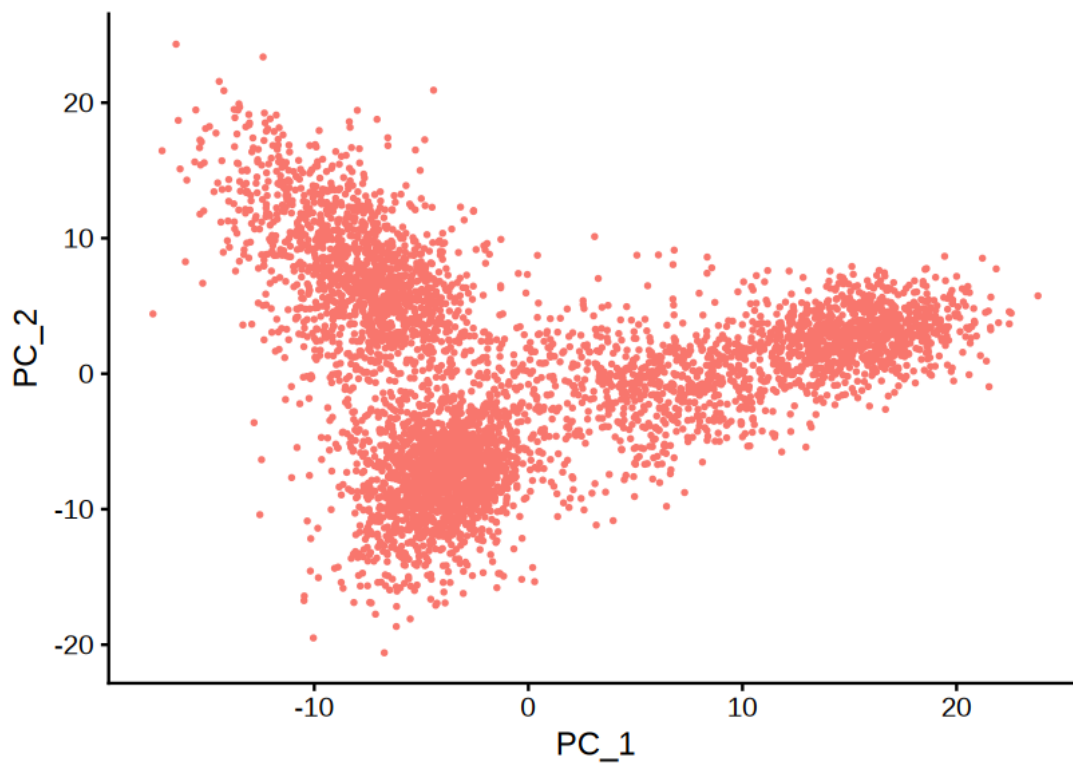
PC_ 5

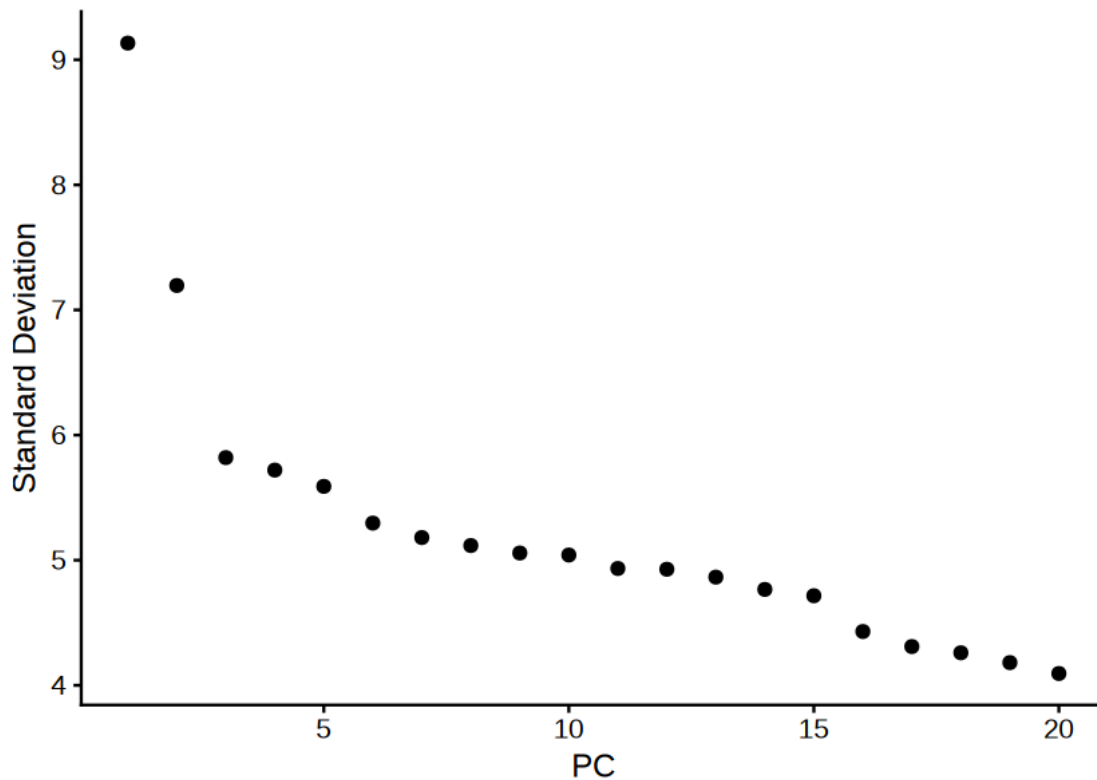
Positive: B2-Mm1a-dup12896, B2-Mm1a-dup8970, B2-Mm1a-dup15634, B2-Mm1t-dup6425, B2-Mm1a-dup7582, B2-Mm1a-dup11368, B2-Mm1a-dup17047, B2-Mm1a-dup2256, B2-Mm1a-dup12636, B2-Mm1a-dup7726

B2-Mm1t-dup3482, B2-Mm1t-dup3308, B2-Mm1a-dup13001, B2-Mm1t-dup12636, B2-Mm1a-dup13612, B2-Mm1a-dup16372, B2-Mm1a-dup10083, B2-Mm1a-dup5102, B2-Mm1t-dup12926, B2-Mm1a-dup12806

B2-Mm1t-dup14884, B2-Mm1a-dup9527, B2-Mm1a-dup15214, B2-Mm1a-dup2700,

B2-Mm1t-dup10228, B2-Mm1t-dup18176, B2-Mm1a-dup14687, B2-Mm1a-dup14446,
B2-Mm1t-dup19058, B2-Mm1a-dup16358
Negative: B2-Mm1t-dup19486, B2-Mm1a-dup14085, B2-Mm1a-dup6804,
B2-Mm1a-dup17735, B2-Mm1a-dup8746, B2-Mm1a-dup9374, B2-Mm1a-dup3576,
B2-Mm1a-dup1213, B2-Mm1a-dup13666, B2-Mm1t-dup311
B2-Mm1a-dup15891, B2-Mm1a-dup11486, B2-Mm1a-dup3991,
B2-Mm1t-dup19367, B2-Mm1a-dup8352, B2-Mm1t-dup3103, B2-Mm1t-dup1923,
B2-Mm1t-dup20804, B2-Mm1a-dup5228, B2-Mm1t-dup19563
B2-Mm1a-dup10034, B2-Mm1a-dup9901, B2-Mm1a-dup14419, B2-Mm1a-dup3896,
B2-Mm1a-dup5923, B2-Mm1t-dup22730, B2-Mm1a-dup16067, B2-Mm1a-dup10698,
B2-Mm1a-dup17808, B2-Mm1a-dup2638





```
[17]: objTE_STARsolo <- FindNeighbors(objTE_STARsolo, dims = 1:10, k.param = 20)
      objTE_STARsolo <- FindClusters(objTE_STARsolo, resolution = 1)
      objTE_STARsolo <- RunUMAP(objTE_STARsolo, dims = 1:10)
      DimPlot(objTE_STARsolo, reduction = "umap")
```

Computing nearest neighbor graph

Computing SNN

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 4790

Number of edges: 150539

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.8735

Number of communities: 25

Elapsed time: 0 seconds

Warning message:

"The default method for RunUMAP has changed from calling Python UMAP via reticulate to the R-native UWOT using the cosine metric"

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|

14:04:33 Writing NN index file to temp file /tmp/RtmpAoRrmn/file3add0dfeafdb5

14:04:33 Searching Annoy index using 1 thread, search_k = 3000

14:04:34 Annoy recall = 100%

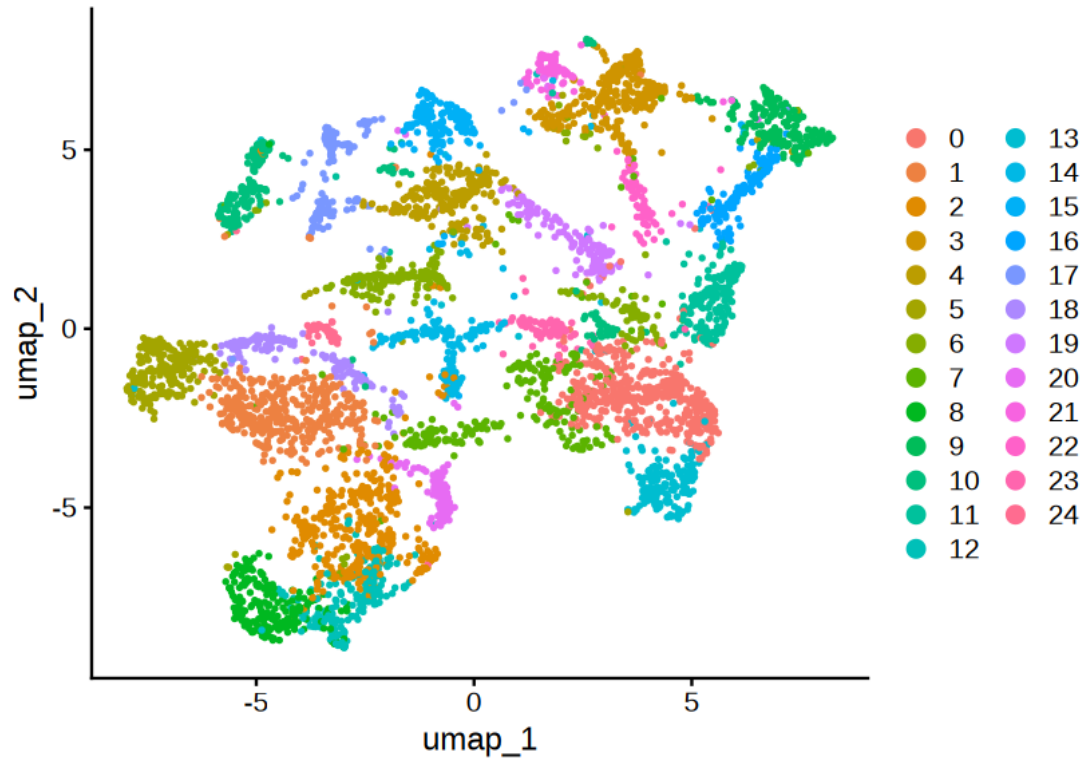
14:04:36 Commencing smooth kNN distance calibration using 1 thread
with target n_neighbors = 30

14:04:39 Initializing from normalized Laplacian + noise (using RSpectra)

14:04:39 Commencing optimization for 500 epochs, with 193196 positive edges

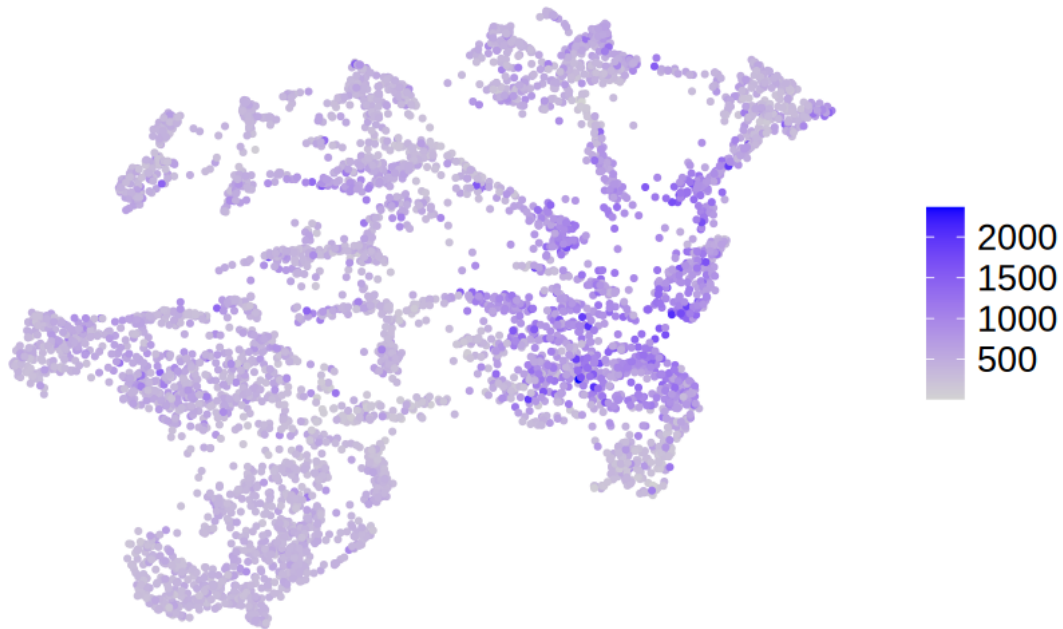
14:04:39 Using rng type: pcg

14:04:47 Optimization finished



```
[18]: FeaturePlot(objTE_STARsolo, reduction = "umap", features = "nCount_TE", pt.size_
      ↪= 0.5) +
      theme_void() +
      theme(text=element_text(size=20))
```

nCount_TE



```
[20]: objTE_STARsolo$celltype <- celltypes$celltype[match(Cells(objTE_STARsolo),  
  ↪celltypes$barcode)]
```

```
[21]: table(objTE_STARsolo$celltype)  
sum(is.na(objTE_STARsolo$celltype))
```

B_cells	Dendritic Cells	Monocytes	Natural Killers	Neutrophils
1601	438	307	90	113
Platelets	T_cells	Unknown		
828	1294	116		

3

```
[26]: mouse_assignment$Barcodes <- sapply(strsplit(mouse_assignment$Barcodes,  
  ↪split="-"), "[", 1)  
table(mouse_assignment$Assignment)  
objTE_STARsolo$Assignment <-  
  ↪mouse_assignment$Assignment[match(Cells(objTE_STARsolo),  
  ↪mouse_assignment$Barcodes)]  
table(objTE_STARsolo$Assignment)
```



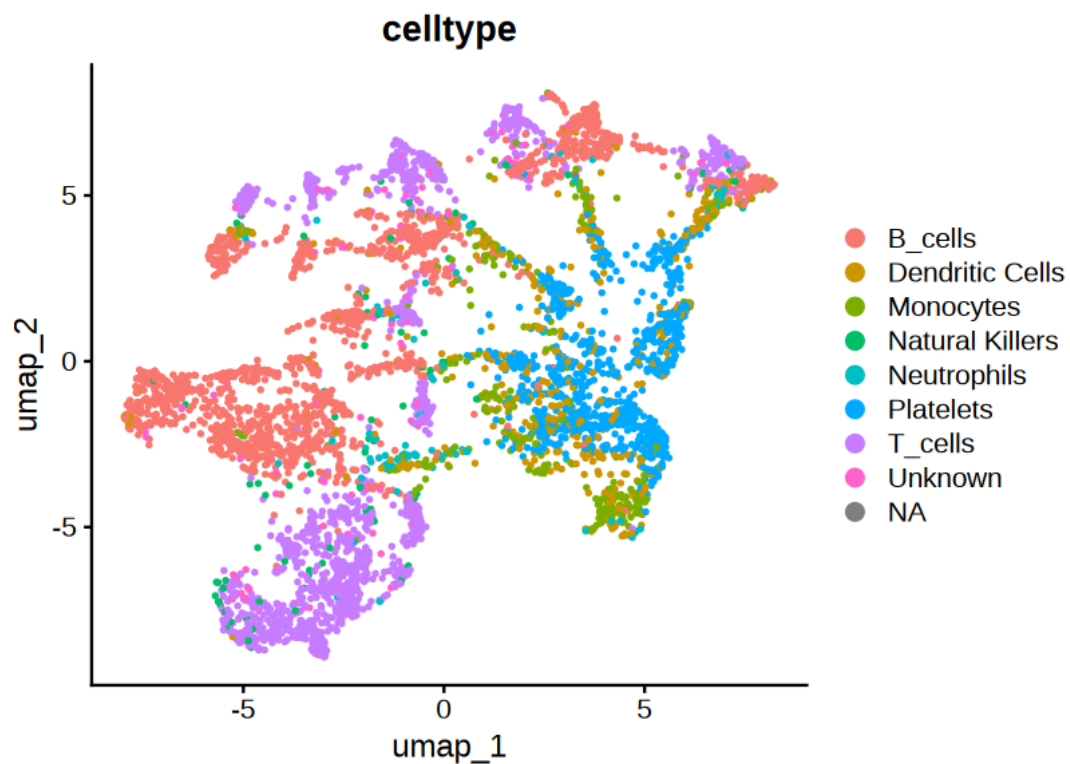
```
# multiplets <- ␣
↪ mouse_assignment$Barcodes[mouse_assignment$Assignment=="Multiplet"]
# filteredBarcodes <- setdiff(filteredBarcodes, multiplets)
```

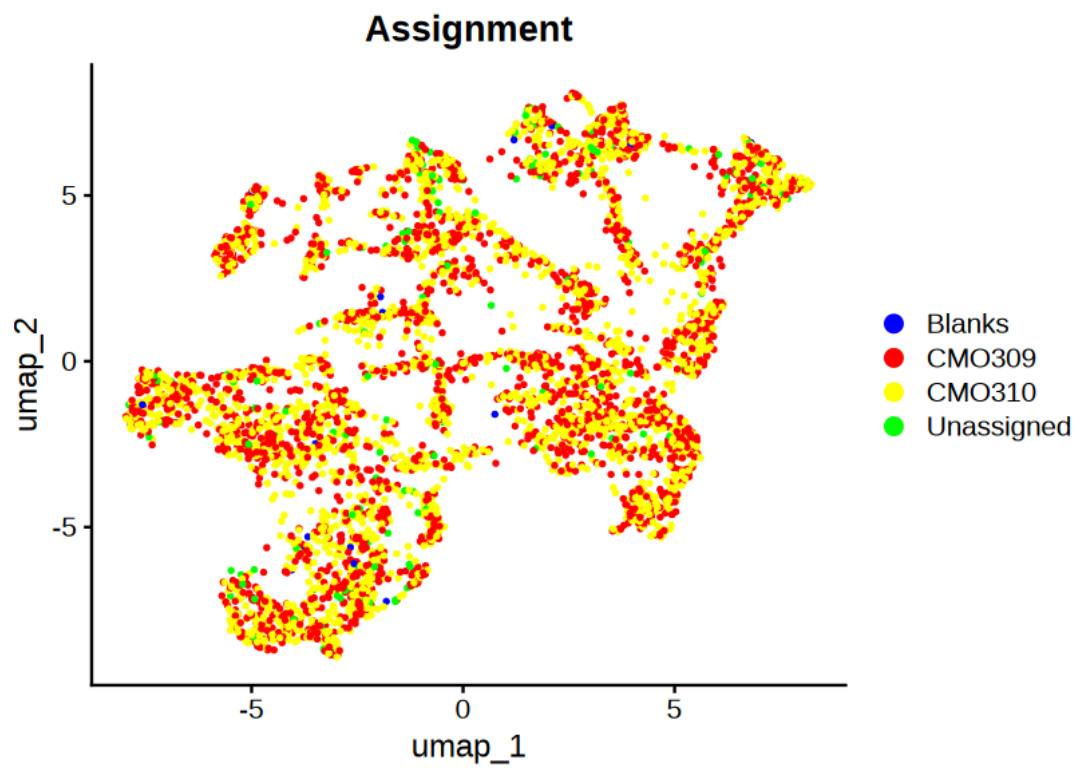
Blanks	CM0309	CM0310	Multiplet	Unassigned
100	4598	5236	253	334

Blanks	CM0309	CM0310	Unassigned
31	2136	2449	174

```
[29]: DimPlot(objTE_STARsolo, reduction = "umap", group.by="celltype")
```

```
DimPlot(objTE_STARsolo, reduction = "umap", group.by="Assignment", ␣
↪ shuffle=TRUE, cols=c("blue","red","yellow","green"))
```





```
[30]: saveRDS(objTE_STARsolo, paste0("data_", dataset_id, "/STARsolo_", dataset_id, "_seuratObj.RDS"))
```

```
[ ]:
```

```
[ ]:
```